

NCERT / DEPFE / CAREER EXHIBITION

Artificial Intelligence Engineer

A career talk for senior secondary students. The pathways, the work, and the future of building intelligent systems.

AUDIENCE

Classes IX to XII

VENUE

KV RK Puram, Sector 2

DEPARTMENT

DEPFE, NCERT, New Delhi

AGENDA

What does a career in AI really involve?

A structured walk through everything you need to decide whether this profession is right for you.

01 Educational Pathway

Subjects, exams and degrees

02 Nature of Work

Day to day tasks and tools

03 Working Conditions and Attributes

Environment and skills that help

04 Career Progression and Salary

Growth from entry to leadership

05 Opportunities and Colleges

Where to work and where to study

06 Courses and Future Outlook

Building employability for the long run

SECTION 01

Getting into the field

There is no single fixed route. A strong grounding in mathematics and programming matters more than any one degree.

01

EDUCATIONAL PATHWAY

How you get into the field

School, Class XI and XII

- **Science stream with Mathematics.** Maths is essential
- Physics and Chemistry for engineering routes
- Opt for **Computer Science** if it is offered
- Start coding early. Python is a great first language

Entrance Exams

- **JEE Main and Advanced** for IITs, NITs, IIITs
- **BITSAT** for BITS Pilani campuses
- **CUET** for central universities
- State CETs and university specific tests

Degree Routes

- **B.Tech or B.E.** in CSE, AI and ML, or Data Science
- **B.Sc** in Computer Science, Maths or Statistics
- **M.Tech, M.S. or PhD** for research and senior roles
- Online degrees and a strong project portfolio also count

What an AI engineer actually does

Core Responsibilities

- Design, train and fine tune **machine learning and deep learning** models
- Collect, clean and prepare data, then engineer features
- Deploy models into products and **monitor them in production**
- Work on NLP, computer vision, recommendations and large language models
- Write reliable, production grade code and run experiments

Tools and Technologies

- **Languages.** Python, SQL, sometimes C++ or Java
- **Frameworks.** PyTorch, TensorFlow, scikit learn, Hugging Face
- **Cloud and DevOps.** AWS, GCP, Azure, Docker, Git
- **Data.** Pandas, NumPy, Spark, vector databases
- Collaborates with data scientists, product and software teams

The environment and what helps you succeed

Hours and Setup

- Typical **40 to 45 hour** work weeks
- Extra hours near launches or deadlines
- **Hybrid and remote** work are common
- On site work for robotics and hardware

Teams and Culture

- Small **cross functional teams**
- Agile sprints, code reviews, stand ups
- Global collaboration across time zones
- Constant need to keep learning

Personal Attributes

- Strong **mathematics** and logical reasoning
- **Curiosity** and self driven learning
- Patience for experimentation and debugging
- Explaining complex ideas **simply**

SECTION 02

Growth and reward

The path runs from intern to leadership. Compensation rises quickly with proven delivery and technical ownership.

02

PROGRESSION AND SALARY

From entry level to leadership

Intern or Trainee

Starting out

Learn on real projects, assist with data and experiments while completing studies.

Junior AI or ML Engineer

0 to 2 years

Build and test models under guidance. Strengthen coding and data skills.

AI or ML Engineer

2 to 5 years

Own features end to end, deploy models to production, mentor juniors.

Senior AI Engineer

5 to 8 years

Lead complex systems, set technical direction, review architecture.

Lead, Principal or ML Architect

8 to 12 years

Design large scale AI systems and guide multiple teams.

Head of AI, Director or CTO

12 plus years

Own AI strategy and organisation wide impact.

ESTIMATED SALARY ALONG THE PATH

Indicative compensation in India

Entry or Junior 0 to 2 years	First roles. Varies widely by employer and city.	6 to 12 LPA
Mid level 2 to 5 years	Proven delivery and production experience.	12 to 25 LPA
Senior 5 to 8 years	Technical ownership and mentoring.	25 to 45 LPA
Lead or Principal 8 to 12 years	Architecture and multi team leadership.	45 to 80 LPA
Director or Head of AI 12 plus years	Strategy. Top product firms pay well above.	80 LPA plus

Figures are broad, indicative annual ranges in Indian Rupees and vary significantly with company, location, skills and global roles. Top product companies and overseas positions can pay considerably more.

Where the demand is

Sectors

- IT and software, e commerce
- Fintech, healthcare, automotive
- Defence, agriculture, telecom
- Edtech, media and entertainment

Organisations

- Global. Google, Microsoft, Amazon, NVIDIA, Meta
- Indian IT. TCS, Infosys, Wipro
- Startups. Sarvam AI, Krutrim and more
- Research. ISRO, DRDO, IITs, IISc

Geographies

- Bengaluru, Hyderabad, Pune
- Gurugram, NCR, Chennai
- Global. US, Europe, Singapore, UAE
- Remote, freelance and consulting

Recommended institutions

Premier Engineering Institutes

- **IITs.** Bombay, Delhi, Madras, Kanpur, Kharagpur, Hyderabad, which has a dedicated AI department
- **IISc Bangalore** for leading AI research
- **NITs.** Trichy, Surathkal, Warangal
- **BITS Pilani.** Pilani, Goa, Hyderabad

Strong CS and AI Programmes

- **IIITs.** Hyderabad, Delhi, Bangalore
- Delhi University, DTU and NSUT
- VIT, Manipal, Ashoka University
- Reputed state and private universities with AI or DS courses

Building employability for the long run

Courses and Certifications

- DeepLearning.AI specialisations by Andrew Ng
- Google, AWS and Azure ML certifications
- Stanford CS231n for vision, CS224n for NLP
- **Kaggle** competitions and Kaggle Learn

Internships and Portfolio

- Research internships at IITs and IISc
- Google Summer of Code and industry internships
- Open source and a **GitHub project portfolio**
- Build things and share them publicly

Future Outlook

- **Generative AI and agents** are reshaping every industry
- Government push through the **IndiaAI Mission**
- Rising focus on **responsible and ethical AI**
- One of the fastest growing, highest demand careers

IN SUMMARY

Build the future, thoughtfully.

Start with strong mathematics and coding, choose a pathway that fits you, keep building projects, and never stop learning. AI rewards curiosity and persistence.

Thank you. Questions and discussion.